Disambiguating “Network” and “Topology” in Paper 3

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Attribution

> *This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026). It does not introduce new axioms. Its contribution is to disambiguate two terms—“network” and “topology”—that appear in Paper 3 with scope-specific meanings, so that downstream work can adopt, extend, or argue against these concepts without ambiguity.*

Abstract

Paper 3 of the CKS trilogy uses “network” in two distinct senses and introduces “topology” as a Claim 6 concept. In the context of specific FAI events (Claims 1–5), “network” is an informal descriptor: the set of Selves currently participating in active FAI relationships at a given time. In Claim 6’s population-scale context, “FAI network” names the emergent pattern of accumulated FAI relationships across many Selves over time—not a platform, not fixed infrastructure, not a centrally administered structure. “Topology” names the configuration of FAI relationships across the network; it emerges from individual Selves’ participation configurations rather than from any central design authority. Both Claim 6 concepts carry calibrated-humility framing: population-scale CKS-governed Self deployments do not yet exist, and network and topology are projections about what the architecture would produce if widely adopted. This note formalizes the disambiguation, states the prior-art coverage each sense provides, and identifies what the disambiguation forecloses for adversarial claims.

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1. Why this disambiguation is needed

Paper 3’s scope extends from the bilateral and small-group FAI event (Claims 1–5) to population-scale collective evolution (Claim 6). This progression introduces a terminological shift: “network” appears in both contexts, but it refers to structurally different objects in each.

In the Claims 1–5 context, “network” is a convenience descriptor for the set of CKS-governed Selves currently participating in FAI relationships—a use that requires no formal definition and carries no structural commitments about persistence, membership, or governance above the bilateral event. In the Claim 6 context, “FAI network” refers to the emergent governance relationship pattern that arises when many Selves independently adopt the FAI architecture and accumulate FAI events over time.

Two misreadings follow from conflating the two senses. The first treats the Claim 6 FAI network as if it were a platform or infrastructure that someone designs and administers—a misreading that would support adversarial claims about novel "AI governance networks" as architectural inventions. The second treats the topology of FAI relationships as a centrally designed property rather than an emergent aggregate of individual governance decisions—a misreading that mischaracterizes the distributed structure of the Claim 6 architecture.

Both misreadings narrow the prior-art record if left uncorrected. This note corrects both.

2. "Network" as informal descriptor (Claims 1–5 context)

In the context of Claims 1 through 5, Paper 3 concerns itself with what happens when two or more CKS-governed Selves form a shared substrate, contribute aspects to it, coordinate under joint authority, handle conflicts through the three-tier mechanism, receive evolution feed through the four-locus mechanism, and dissolve the shared substrate after the FAI event concludes.

The Selves involved in this process are, while the event is active, in an FAI relationship with each other. When Paper 3 refers to "the network" in this context, it means the set of Selves currently connected by active FAI relationships at a given moment. This is a descriptive use: it names current participants without implying fixed membership, persistent infrastructure, or a formal governance structure spanning those participants beyond the event itself.

The network in this sense is as transient as the FAI events that constitute it. It expands when new FAI events begin, contracts when events conclude, and has no persistent form between events. It does not require a central registry, a coordination layer above the participating Selves, or any governance object that outlasts the event that produced it. It is shorthand for "the FAI-participating Selves at this time."

This informal sense is adequate for Claims 1–5 because those claims concern the mechanics of a specific FAI event—not what accumulates across many such events. The informal descriptor requires no further formal definition at this scope.

3. "FAI Network" as emergent relationship pattern (Claim 6 scope)

Claim 6 extends Paper 3's architecture to population scope. When FAI events accumulate across many CKS-governed Selves—across diverse domains, owned by different organizations, governed under distinct authority structures—they constitute a collective evolution mechanism at population scope. The "FAI network" at Claim 6 scope is the emergent pattern that this accumulation produces. Several properties distinguish it from the informal descriptor of §2.

Emergence, not design. No one designs or administers the FAI network at Claim 6 scope. It is what comes to exist when many Selves independently adopt the FAI architecture and form bilateral or small-group governance relationships. The network's existence depends on adoption; its properties depend on the governance configurations each participating Self brings; its scale depends on how many Selves adopt the architecture over time. None of these dependencies point to a central authority that plans or administers the network.

Not fixed infrastructure. The FAI network is not a platform with persistent nodes, registered participants, and a defined topology assigned at deployment time. It is a governance relationship pattern—the cumulative imprint of many FAI events across many Selves. Its configuration at any moment is the aggregate of which Selves are currently engaged in FAI relationships and what governance configurations govern those relationships. Between FAI events, there is no infrastructure to maintain.

No central governance authority. Each Self governs its own FAI relationships through its participation configuration. No Self governs another Self's participation decisions. No authority above the individual Self level determines who participates, with whom, or under what terms. The absence of central governance authority is not a gap the architecture leaves to be filled; it is a structural commitment of the design. Population-level governance properties emerge from individual Selves' decisions, not from a governance layer positioned above them.

Scale is indeterminate. How large the FAI network becomes, how densely connected it grows, and what population-level dynamics it exhibits are projections contingent on adoption at scale. Claim 6 projects convergence, capability propagation, and specialization-preservation as architectural consequences of widespread FAI adoption; it does not guarantee these as outcomes. The calibrated-humility register is appropriate here: population-scale CKS-governed Self deployments do not yet exist, and the network's empirical properties await observation at a scale that has not yet been reached.

The Claim 6 FAI network is, at its core, an emergent governance phenomenon: it is what exists when the FAI architecture is widely adopted. It is not something anyone builds in advance.

4. Topology: aggregate of individual participation configurations

"Topology" at Claim 6 scope names the configuration of FAI relationships across the network—specifically, which Selves have which FAI relationships with which other Selves. Each Self's participation configuration specifies its local topology: which other Selves it will engage with in FAI events, under what governance conditions, and with what sharing scope. The aggregate of all participating Selves' participation configurations constitutes the network topology.

Topology at Claim 6 scope is not centrally designed. No authority specifies the global topology in advance or maintains it as a managed artifact. Instead, topology emerges from the accumulation of individual governance decisions made independently by each participating Self. Topology governance is therefore distributed: each Self governs its own local topology contribution, and no authority governs the aggregate topology as a whole.

Two consequences follow from this structure. First, the network topology is heterogeneous by construction. Different Selves bring different participation configurations—different sharing scopes, different partner preferences, different governance conditions for FAI engagement. The aggregate topology reflects this diversity. Common patterns may propagate where governance configurations enable propagation; specializations are preserved where governance configurations protect them. The topology is not uniform and is not expected to converge toward uniformity.

Second, the topology evolves without central redesign. As Selves update their participation configurations, as new Selves join the network, or as existing Selves dissolve FAI relationships, the aggregate topology shifts through the accumulation of individual governance decisions—each Self acting within its own governance perimeter—rather than through a centrally managed reconfiguration.

This distributed governance architecture for topology is one of the structural features that distinguishes the Claim 6 FAI network from coordinator-mediated federation architectures, where topology is typically assigned at the consortium or platform layer with participants joining as registered nodes in a pre-defined structure. The Claim 6 architecture commits to the opposite: topology is a governance consequence, not a governance input.

5. Prior-art coverage: what this disambiguation forecloses

The two-sense disambiguation produces three pieces of prior-art coverage, each foreclosing a distinct adversarial claim.

Coverage 1—The informal network sense (§2). The prior art covers treating a set of CKS-governed Selves currently engaged in FAI relationships as a named coordination object. Any claim to novel invention in the concept of "a set of AI governance participants connected by shared governance operations" must address that Paper 3 employs this concept as a working descriptor for Claims 1–5, establishing it in the prior-art record before the adversarial claim.

Coverage 2—The emergent FAI network sense (§3). The prior art covers the Claim 6 network as an emergent governance relationship pattern arising from independent adoption by many Selves. Adversarial claims about novel "AI governance networks" as architectural inventions must address that the CKS FAI network is not a platform or infrastructure to be built, but an emergent consequence of individual Self governance decisions. A claim to a novel "AI governance network platform" would need to argue that a centrally administered platform is architecturally distinct from the CKS emergent network—not that the network concept itself is new.

Coverage 3—Topology as distributed aggregate (§4). The prior art covers distributed topology governance in AI coordination architectures, where individual participants specify their own coordination relationship configurations and the aggregate of those specifications constitutes the system topology. Any claim to novel invention in this structure—whether framed as "participant-governed topology," "decentralized relationship configuration," or related constructions—must address Paper 3's explicit Claim 6 commitment to topology as an aggregate of individual participation configurations rather than a centrally designed property.

6. Calibrated-humility anchor

Both Claim 6 concepts—the emergent FAI network and its topology—require calibrated-humility framing, and the prior-art record benefits from naming this explicitly.

Paper 3's population-scale claims are projections about what the CKS architecture would produce under widespread adoption, not empirical observations of an existing deployment. Population-scale

CKS-governed Self deployments do not yet exist. The specific dynamics of the FAI network—convergence rates, topology stability, capability propagation patterns across organizational and domain boundaries—are downstream of empirical observation at a scale that has not yet been reached. Future work on formal protocol specifications, deployment architectures, and empirical validation at scale will refine the projections; the projections themselves are not the claim.

The calibrated-humility framing does not weaken the prior-art coverage. Prior art does not require empirical validation of the claimed architecture; it requires disclosure of the architectural commitment. Paper 3's Claim 6 discloses the commitment—FAI events accumulating across many Selves produce an emergent governance relationship network whose topology emerges from individual governance decisions—and this note formalizes that disclosure in operational terms.

The uncertainty in the Claim 6 projections is specifically about scale and observed dynamics: how many Selves, how quickly, producing what observable emergent properties. On the architectural structure—network as emergent pattern, topology as distributed aggregate, no central administration layer—Paper 3 is architecturally precise. That structural commitment is what the prior-art record carries forward, and what this note formalizes.

Source papers

Li, W. (2026). Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). Inter-Self Coordination via Shared Substrate / Full Aspect Integration. April 2026. ORCID: 0009-0004-8065-3235.

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